

CSA0904 – PROGRAMMING FOR JAVA

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C03 – AT1

1. Debug

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar indicates the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1093`. The page header includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" and "JAVA PROGRAMMING". The user's name "Dharshan A N" and registration number "192525370RA" are displayed in the top right corner.

The code editor shows the following Java code:

```
1 // ...
2
3 producer.start();
4 consumer.start();
5
6 producer.join();
7 consumer.join();
8
9 System.out.println("Queue size: " + queue.size());
10 }
11 }
```

The "Your Code" section shows the full code for the `ProducerConsumerDemo` class:

```
1 import java.util.concurrent.*;
2
3 public class ProducerConsumerDemo {
4     public static void main(String[] args) throws InterruptedException {
5
6         BlockingQueue<Integer> queue = new ArrayBlockingQueue<>(5);
7
8         Thread producer = new Thread() -> {
9             for (int i = 0; i < 10; i++) {
10                 queue.add(i);
11             }
12         };
13
14         Thread consumer = new Thread() -> {
```

The "Input" section shows "stdin input". The "Output" section shows the following error message:

```
Queue size: 5
read "Thread-1" Exception in
thread "Thread-0"
java.lang.IllegalStateException:
Queue full
```

2. Debug

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The code editor shows the following Java code:

```
1 public class SynchronizedCounterDemo {
2     private int count = 0;
3
4     public void increment() {
5         count++;
6     }
7
8     public synchronized void decrement() {
9         count = count - 1;
10    }
11
12    public static void main(String[] args) throws InterruptedException {
13
14        SynchronizedCounterDemo demo = new SynchronizedCounterDemo();
15
16        Runnable task = () -> {
17            for (int i = 0; i < 1000; i++) {
18                demo.increment();
19            }
20        };
21
22        Thread t1 = new Thread(task);
23        Thread t2 = new Thread(task);
24
25        t1.start();
26        t2.start();
27    }
28 }
```

The "Input" section shows "stdin input". The "Output" section shows the following output:

```
Final count: 1711
1711
```

3. Debug

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1093". The lab header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and a user profile for "Dharshan A N" with ID "192525370RA".

On the left, a green box indicates a question status: "Q4 Finally Block Overriding Return Values 25 marks 4/4 answered".

The main workspace contains the following code and output:

```
int result = future.get(); caller context
System.out.println(result);

Runnable r = () -> {
    return 5;
    executor.submit(r);
    executor.shutdown();
}
```

Your Code

```
1 import java.util.concurrent.*;
2
3 public class RunnableCallableDemo {
4     public static void main(String[] args) throws Exception {
5
6         ExecutorService executor = Executors.newFixedThreadPool(2);
7
8         Callable<Integer> callableTask = () -> {
9             return 10 / 0;
10        };
11
12        Future<Integer> future = executor.submit(callableTask);
13
14        try {
```

Input

stdin input

Output

```
Exception:
java.lang.ArithmeticException: / by
zero
Runnable task executed
```

The Windows taskbar at the bottom shows a temperature of 90°F, "Partly cloudy" weather, and system time of 19:46 on 13-08-2026.

4. Debug

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1093". The lab header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and a user profile for "Dharshan A N" with ID "192525370RA".

The main workspace contains the following code and output:

```
System.out.println(y); might not be initialized
}

static int compute() {
    return 42;
}

}
```

Your Code

```
1 public class FinallyReturnDemo {
2
3     public static int riskyDivision(int a, int b) {
4         try {
5             int result = a / b;
6             return result;
7         } catch (ArithmeticException e) {
8             return -1;
9         } finally {
10            System.out.println("Finally executed");
11        }
12    }
13
14    public static void main(String[] args) {
15
16        System.out.println(riskyDivision(10, 0));
17    }
```

Input

stdin input

Output

```
Finally executed
-1
Caught: fall
x = 20
Finally executed
y = 42
```

The Windows taskbar at the bottom shows a temperature of 90°F, "Partly cloudy" weather, and system time of 19:46 on 13-08-2026.